Akoizo Original Practice - Form 1 - Answer key

Division C | Forensics | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Which statement best expresses Locard exchange?

Unit: Forensic Method and Locard Exchange.
Worked-case data: A blue polyester fiber on a chair resembles fibers from a suspect’s common jacket. The jacket was present in the room during an unrelated earlier meeting.
Separate investigation observations:
Material: A common blue polyester fiber is recovered.
Comparison: Several garments in the reference set share the same observed color and polymer class.
Context: One matching garment was in the room before the incident during ordinary use.

A. Every contact leaves a trace

B. Fingerprints are unique

C. DNA proves guilt alone

D. Confirmatory tests never err

Correct: A

2. (4 points) Why run a distilled-water blank with Benedict reagent?

Unit: Lab Safety, QA, and Documentation.
Worked-case data: A reagent blank and an unknown both turn yellow. The expected negative blank should remain clear.
Separate investigation observations:
Positive control: The known target gives the expected response.
Reagent blank: The blank also gives a strong response that should not occur.
Unknown: The unknown produces the same response.

answer: To show the reagent background blue color so a negative is not mistaken for a weak positive. (4 points).

3. (2 points) Powder fizzes with HCl and gives basic pH. Best inference?

Unit: Powder Scheme: Solubility, pH, and Precipitation.
Worked-case data: The exercise candidate set is sodium chloride, sodium carbonate, and calcium carbonate. The unknown reacts with acid to release gas and is poorly soluble in water.
Separate investigation observations:
Water solubility: Most of the white solid remains undissolved.
Acid response: A fresh aliquot releases gas; the blank remains negative.
Reference comparison: In this candidate set, calcium carbonate is poorly soluble and acid-reactive; sodium carbonate is soluble; sodium chloride is not acid-effervescent.

A. Sucrose

B. Starch

C. Chloride salt

D. Carbonate

Correct: D

4. (2 points) Every sample and the tool blank show yellow emission. Best next step?

A. Mix all unknowns

B. Accept every sample as pure sodium salt

C. Resolve carryover and obtain a clean blank

D. Ignore the blank

Correct: C

5. (2 points) Which polymer is a condensation product?

Unit: Polymer Chemistry and Recycling Codes.
Worked-case data: A fragment floats in water. The reference set contains polypropylene and two polyethylene samples with densities below water.

A. HDPE

B. PET

C. PP

D. PS

Correct: B

6. (2 points) Green Beilstein flash means:

Unit: Plastic ID: Density, Flame, and Beilstein.
Worked-case data: A fragment sinks in a 0.92 g/mL reference liquid and floats in a 0.98 g/mL reference liquid. A candidate table lists 0.90, 0.95, and 1.20 g/mL samples.

A. Halogen such as PVC chlorine

B. Starch

C. Glucose

D. Sodium

Correct: A

7. (2 points) Twisted ribbon, paper odor, soft ash. Fiber?

Unit: Fibers: Classification and Microscopy.
Worked-case data: The questioned fiber is a flattened twisted ribbon with a cellulose-associated reference response. Reference A is cotton, B is wool, and C is a uniform synthetic filament.
Separate investigation observations:
Longitudinal view: A flattened ribbon with repeated twists is visible.
Composition record: The supplied reference assay supports a cellulose-based material.
Comparison set: Wool shows scales; the synthetic reference is a smooth uniform filament; cotton shows convolutions.

A. Nylon

B. Polyester

C. Wool

D. Cotton

Correct: D

8. (4 points) Medulla width 40 um, shaft 60 um. Index and inference?

answer: Index 0.67, above 0.5, consistent with animal hair rather than human. (4 points).

9. (4 points) Spot 4.2 cm, front 7.0 cm. Rf?

answer: Rf equals 4.2 divided by 7.0 equals 0.60. (4 points).

10. (2 points) Tallest peak in a mass spectrum is called:

Unit: Mass Spectrometry and Spectroscopy.
Worked-case data: A spectrum contains a strong m/z 91 fragment. Candidate A and candidate B can both produce that fragment, but their molecular masses differ.
Separate investigation observations:
Base peak: The largest signal is m/z 91.
Candidate comparison: Two candidates in the supplied reference set produce m/z 91.
Mass constraint: The intact-ion region is too weak to interpret reliably in this record.

A. Base peak

B. M+2

C. Solvent front

D. Molecular ion

Correct: A

11. (2 points) Loop vs whorl delta count?

Unit: Prints: Patterns and Minutiae.
Worked-case data: A questioned print and a reference both show a loop. The questioned image has a large smear through the region where detailed features would be compared.
Separate investigation observations:
Ridge flow: Both images show a loop pattern.
Detail quality: Most of the questioned print is smeared; only two short clear segments remain.
Candidate context: Several reference prints share the same pattern class.

A. Both 0

B. Both 2

C. Loop 1, whorl 2

D. Loop 2, whorl 1

Correct: C

12. (2 points) What should guide method selection for a paper note?

Unit: Developing Latent Prints.
Worked-case data: Exhibit A is an uncoated paper note. Exhibit B is a smooth plastic container. Both may have latent residues.
Separate investigation observations:
Surface: The exhibit is uncoated absorbent paper.
Residue: The question concerns amino-acid-containing latent residue.
Other evidence: The note also has ink that may require separate comparison.

A. The darkest reagent

B. Porosity, residue chemistry, and other evidence needs

C. A universal method order for every surface

D. The suspect’s identity

Correct: B

13. (2 points) Heterozygote at one STR locus shows:

Unit: DNA: STR Profiling and Electrophoresis.
Worked-case data: The questioned profile is locus 1: 10,12 and locus 2: 8,9. Reference A is 10,12 and 8,9. Reference B is 10,11 and 8,9.
Separate investigation observations:
Questioned profile: Locus 1: 10,12. Locus 2: 8,9.
Reference profiles: A: 10,12 / 8,9. B: 10,11 / 8,9.
Quality statement: This teaching dataset is complete, uncontaminated, and single-source; no dropout is modeled.

A. One peak

B. Two peaks

C. No peaks

D. Three peaks

Correct: B

14. (2 points) Parents AO x BO can have child type:

Unit: Blood: ABO Genetics and Spatter.
Worked-case data: A stain has a 6 mm minor axis and a 12 mm major axis. Its valid typing reacts only with anti-A.

A. A, B, AB, or O

B. AB only

C. B only

D. O only

Correct: A

15. (2 points) Radial cracks make right angles on which side?

Unit: Glass Fracture and Ballistics.
Worked-case data: A questioned fragment measures RI 1.520 ± 0.002. Reference A is 1.521 ± 0.002; reference B is 1.535 ± 0.002 under comparable conditions.
Separate investigation observations:
Questioned fragment: RI 1.520 ± 0.002.
Reference A: RI 1.521 ± 0.002 under the same stated conditions.
Reference B: RI 1.535 ± 0.002 under the same stated conditions.

A. Both equally

B. Neither

C. Impact side

D. Reverse from impact

Correct: D

16. (4 points) 24 h at 22 C with base 6 C gives ADH?

answer: 16 times 24 equals 384 degree-hours. (4 points).

17. (4 points) Rewrite as CER: powder is CaCO3.

answer: Claim CaCO3 consistent; evidence HCl fizz plus brick-red flame plus basic pH; reason carbonate plus calcium chemistry with negative Benedict ruling out sugars. (4 points).

18. (4 points) Why can one reliable exclusion outweigh several class matches?

Unit: Mock Crime Synthesis and Time Plan.
Worked-case data: Candidate A's pen is compatible with a questioned ink pattern. A complete single-source stain profile excludes A as its donor. The scenario does not state that the writer and stain donor must be the same person.
Separate investigation observations:
Ink comparison: A's pen and the questioned ink have compatible developed patterns.
DNA comparison: A is excluded as sole donor of a separate clean single-source stain.
Scenario constraint: The case does not require the writer and stain donor to be the same person.

answer: Exclusion breaks single-source hypothesis while common class matches occur by chance across people. (4 points).